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| **PI Name:** | MacNeilage, Paul R. Ph.D. | | |
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| **Zip Code:** | 63110 | **Congressional District:** | 1 |
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| **COI Name (Institution):** | Angelaki, Dora ( MENTOR/Washington University ) | | |
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POSTDOCTORAL FELLOWSHIP

Spatial orientation is achieved by integrating sensory information from different modalities in order to estimate both body orientation and self-motion. Multiple sources of relevant sensory information are often available, so the question becomes, how should the nervous system combine them? If the goal is to achieve the single most probable combined estimate, then the rule is simple; each information source should be weighted based on its relative reliability. This kind of statistically optimal combination rule is known as Bayesian estimation, and it has been shown to accurately characterize perception in a variety of situations.

One aim of our research was to develop a comprehensive Bayesian model for spatial orientation perception. In order to develop the best possible model, we conducted a thorough review of existing spatial orientation models (MacNeilage et al. 2008). We focused on statistically optimal models of spatial orientation that are dynamic, meaning that the model input and output are continuous and vary over time. We concluded that the particle filter technique is best suited to modeling perception of spatial orientation because it is statistically optimal, the architecture is distributed rather than assuming particular feedback circuits, and most importantly, it is capable of implementing the non-linear system dynamics that are required for spatial orientation computations.

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| **Task Description:** | Our review has provided key insights that will guide future modeling efforts. We have discussed these issues in some detail with other NSBRI investigators developing online systems that can predict spatial disorientation and alert crew members in situations when it is likely to occur (Project title: "Modeling and Mitigating Spatial Disorientation in Low-Gravity Environments"; Principal investigator: Ronald L. Small).<br><br>The second aim of our research was to measure visual and vestibular thresholds for detecting and discriminating spatial orientation stimuli. We conducted three such experiments using a motion simulator with attached stereo visual display. We used a two-alternative-forced-choice procedure because this method minimizes potential sources of response noise and bias. Also, because all data were collected using a common method and apparatus, it is possible to make interesting comparisons across conditions.<br><br>The first experiment investigated perception of heading, which is the direction of self-motion. Heading perception is fundamental to effective navigation and vehicle guidance, so it is important to understand the factors that influence the precision of visual and vestibular heading estimates. We measured discrimination thresholds for heading azimuth and elevation in visual-only and vestibular-only conditions with observers oriented upright and side-down relative to gravity. Visual thresholds were significantly lower than vestibular, and upright thresholds were generally lower than side-down. Vestibular results revealed that observers are better at discriminating head-centric azimuth than elevation, regardless of body orientation. In other words, sensitivity to heading depends more upon the direction of self-motion relative to the head, than relative to gravity. We conducted two follow-up experiments using the same subjects. We found that performance on the three tasks was correlated and determined that the asymmetric vestibular sensitivity to heading stems from an underlying difference in sensitivity to linear acceleration along the interaural and dorsoventral axes of the head.<br><br>In a second experiment we investigated the question of whether or not vestibular cues to linear and angular self-motion interact during rotation about an earth-vertical axis. Linear and angular self-motion is signaled by the otoliths and canals, respectively. During rotation about an earth-horizontal axis (i.e. tilt), these signals must interact so that the brain correctly interprets the change in the otolith signal as due to a change in the direction of gravity. In contrast, we found no evidence of canal-otolith interaction during rotation about an earth-vertical axis. Thresholds for detecting linear acceleration were not influenced by the speed of angular motion.<br><br>In a third experiment, we investigated whether or not vestibular cues to linear self-motion facilitate the detection of visual object motion during self-motion. Linear self-motion gives rise to a distinctive globally consistent pattern of motion on the retina known as optic flow. Independently moving objects in the scene will generate visual motion signals that deviate from the global pattern. Thus, observers must parse the optic flow in order to judge object motion during self-motion. We found that thresholds for discriminating object motion in optic flow patterns were reduced when vestibular signals consistent with the linear self-motion depicted by the optic flow were presented simultaneously. In other words, vestibular signals facilitate optic flow parsing.<br><br>The research described above contributes to a better understanding of spatial orientation in general. The modeling review provides unique insights that will be valuable for developing and evaluating models to predict and investigate situations where spatial disorientation is likely to occur. The results of the experiments provide a better understanding of visual and vestibular perception of spatial orientation stimuli which may be applied to the development of countermeasures like better cockpit display technology, and improved motion simulations. The psychophysical methods employed could be used to develop novel techniques for crew training and evaluation. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | Spatial orientation allows astronauts to pilot the space shuttle and navigate their way around the inside of the international space station. It also allows people here on earth to get where they are going and avoid collisions while driving a car or riding a bike. Perception of spatial orientation is essential for mobile organisms to navigate effectively within the environment. Without perception of spatial orientation we are literally lost. By successfully modeling perception of spatial orientation we can predict when spatial disorientation is most likely to occur. This will help astronauts avoid potentially disorienting and hazardous situations in space. It could also help avoid spatial disorientation here on earth. For example, it could lead to the development of technology that helps drivers avoid disorientation and accidents on the road.<br>The psychophysical measurements we are making also contribute to a better general understanding of spatial orientation perception. The methodologies we have developed for these experiments could be used to assess performance of spatial orientation tasks by astronauts. They could also be used to assess the performance of aircraft pilots or patients with visual or vestibular deficits. |
| **Task Progress:** | We have evaluated existing dynamic models for perception of spatial orientation (see MacNeilage, Ganesan, & Angelaki, 2008). The formerly proposed Bayesian model (MacNeilage et al., 2007) is limited to static situations. We simulated and tested dynamic models proposed by Borah et al. (1988) and Laurens & Droulez (2007) which use Kalman and Particle filter techniques, respectively. We conclude that the particle filter technique is best suited to modeling perception of spatial orientation because it is dynamic, statistically optimal, and capable of incorporating the non-linear system dynamics that are integral to spatial orientation computations.<br>We used two-interval-forced-choice experiments to measure visual and vestibular sensitivity to spatial orientation stimuli. In particular, we measured thresholds for discriminating heading, which is the direction of linear self-motion, and investigated the dependency of discrimination performance on movement direction in head-coordinates and world-coordinates. We related vestibular heading thresholds to baseline measures of sensitivity to linear acceleration in head and world coordinates. In another experiment we determined that canal and otolith signals do not interact in subjects asked to detect linear acceleration during angular movement. These threshold measures may be used to determine the necessary parameters of Kalman or particle filter models described above, which depend on the reliability or noise present in vestibular sensory signals. In addition, the experimental methods we developed could be used to assess sensitivity of crew members to spatial orientation stimuli more precisely than is possible with standard medical diagnostics.<br><br>A final experiment investigated vestibular contributions to discriminating object motion during self-motion. This task requires observes to parse the visual optic flow pattern in order to estimate object motion and self-motion separately. Thresholds were lower when congruent vestibular stimulation was presented. This is one more demonstration of how visual and vestibular self-motion signals are intimately linked and should be considered in tandem as different parts of the same problem. |
| **Bibliography Type:** | Description: (Last Updated: 02/11/2010) |
| **Abstracts for Journals and Proceedings** | MacNeilage PR, Angelaki DE. "Vestibular heading discrimination in humans is better for head-centric azimuth than elevation regardless of body orientation." Neuroscience 2008, Society for Neuroscience Meeting, Washington, DC, November 15-19, 2008.<br>Program No. 18.5. Society for Neuroscience, Online, 2008.<br>http://www.abstractsonline.com/Plan/ViewAbstract.aspx?sKey=36c5eea2-17b9-4a8f-877c-539dd9729ce3&cKey=cd1e01f7-0dc7-42bc-9483-71410ec388d0 , Nov-2008 |
| **Abstracts for Journals and Proceedings** | MacNeilage P, Angelaki D. "Visual and vestibular discrimination of heading azimuth and elevation for upright and side-down observers." Vision Sciences Society Meeting, Naples, Florida, May 9-14, 2008.<br>J Vision. 2008;8(6):834a. http://journalofvision.org/8/6/834/ , May-2008 |
| **Abstracts for Journals and Proceedings** | Turner AH, MacNeilage PR, Angelaki DE. "Absence of canal-otolith interaction during detection of linear component of curved-path self-motion." Neuroscience 2008, Society for Neuroscience Meeting, Washington, DC, November 15-19, 2008.<br>Program No. 367.16. Society for Neuroscience, Online, 2008.<br>http://www.abstractsonline.com/Plan/ViewAbstract.aspx?sKey=a4102a70-1b51-403d-97fd-1736b6374cac&cKey=1bfd1d10-de47-4d8c-9e49-969981f3d92b , Nov-2008 |

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| **Abstracts for Journals and Proceedings** | Zhang Z, MacNeilage PR, Angelaki DE. "Vestibular facilitation of visual motion segmentation; a role for incongruent visual-vestibular MSTd neurons?" Neuroscience 2008, Society for Neuroscience Meeting, Washington, DC, November 15-19, 2008. Program No. 461.3. Society for Neuroscience, Online, 2008. http://www.abstractsonline.com/Plan/ViewAbstract.aspx?sKey=77ff31a5-583c-425f-8a6c-8667bebe75c4&cKey=e8c7f3fc-01b4-4654-a6df-38808da91fb5 , Nov-2008 |
| **Articles in Peer-reviewed Journals** | MacNeilage PR, Ganesan N, Angelaki DE. "Computational approaches to spatial orientation: from transfer functions to dynamic Bayesian inference." J Neurophysiol. 2008 Dec;100(6):2981-96. PubMed PMID: 18842952 , Dec-2008 |